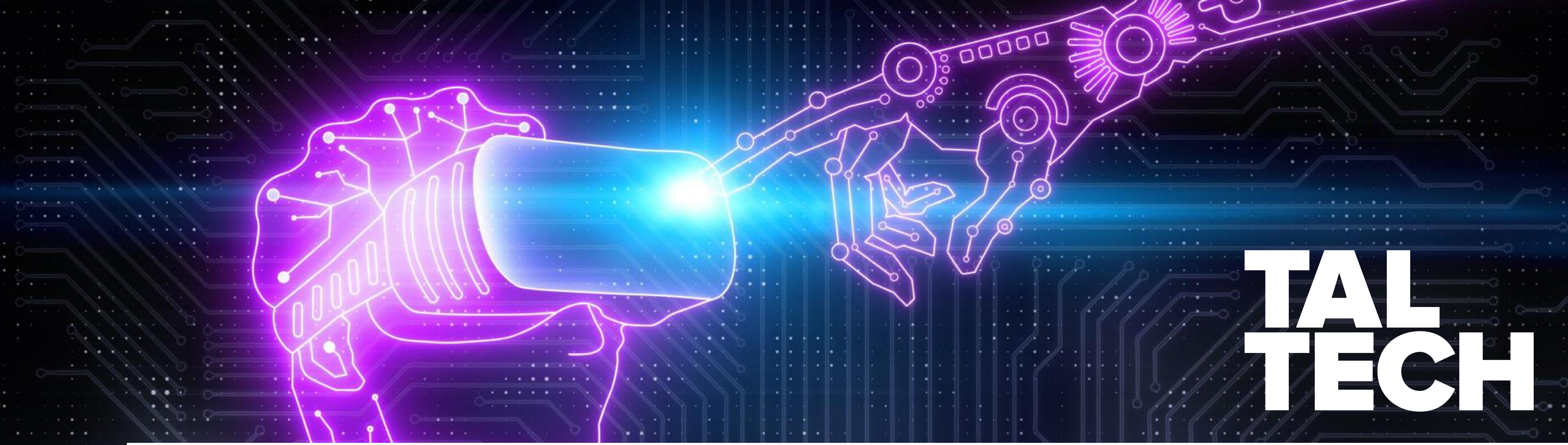

ITE4110: INFORMATION SOCIETY PRINCIPLES: TOWARDS E-GOVERNANCE

DR. ERIC JACKSON, JOSEPHINE LUSI

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E-GOVERNANCE TECHNOLOGIES AND SERVICES MASTER'S PROGRAM

QUICK COURSE ROUNDUP

- **Today's Agenda**
  - Data Governance/Open Data/AI
  - Guest Lecturer: Gledson Pompeu, Federal District, Brazil · Advisor on Innovation and Digital Government · Tribunal de Contas da União
- **Task 3-5 Feedback**
  - Task 4: feedback is done, will be provided after the class
  - Task 5: Will be provided by the end of the week
- **Quick Reflection: How has the case sign-up process been?**

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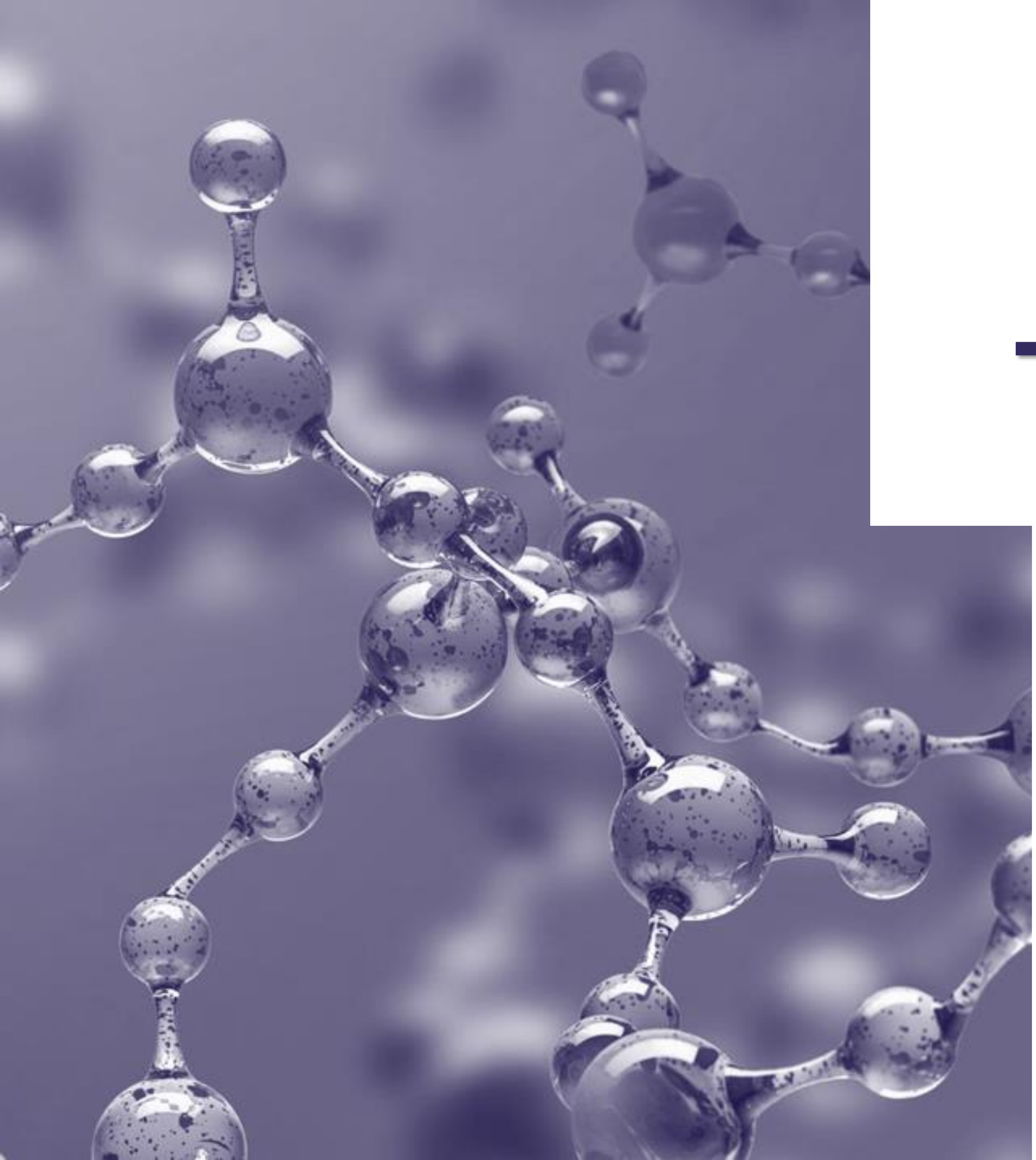

THE BUILDING BLOCKS OF DATA GOVERNANCE

WHAT IS DATA GOVERNANCE? FROM THE LITERATURE

- Explosion of data volumes and different data types as well as Big Data
- "Data governance is the exercise of authority and control over the management of data"
- Fundamental to the public sector, large enterprises, and even SMEs in the digital age
- In theory, data governance should be engineered towards maximizing the value of data as an asset
- Regulatory environment, especially in the EU, has made data governance a compliance issue as well (GDPR, Data Act, Data Governance Act, etc.)
- Has become a primary agenda issue for developing nations who are seeking to digitally transform

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DATA GOVERNANCE DEFINITION IN THE LITERATURE

“Data governance specifies a cross-functional framework for managing data as a strategic enterprise asset. In doing so, data governance specifies decision rights and accountabilities for an organization’s decision-making about its data. Furthermore, data governance formalizes data policies, standards, and procedures and monitors compliance.”

Data governance: A conceptual framework, structured review, and research agenda. Rene Abraham, Johannes Schneider, Jan vom Brocke (2019)

- Looked into 145 data governance studies from 2001-2019
- Did not assess Big Data Governance per se
- Provide a holistic data governance framework
- Most cited data governance article in recent literature (700+)

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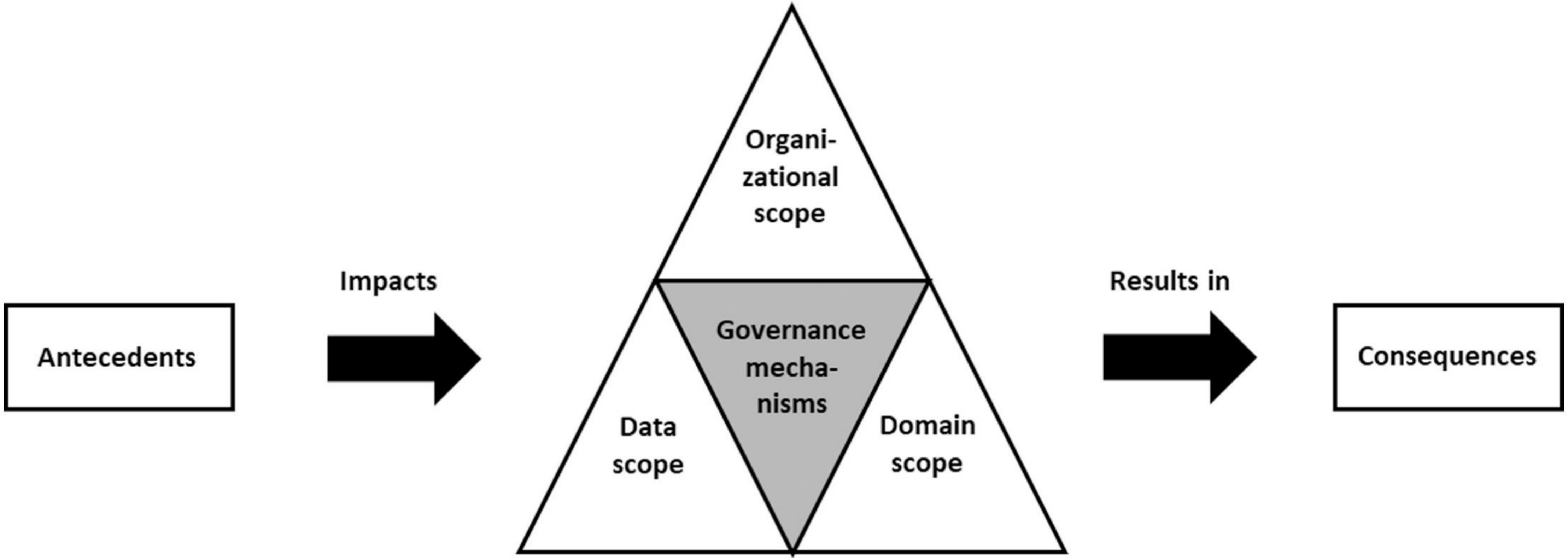
DATA GOVERNANCE FRAMEWORK



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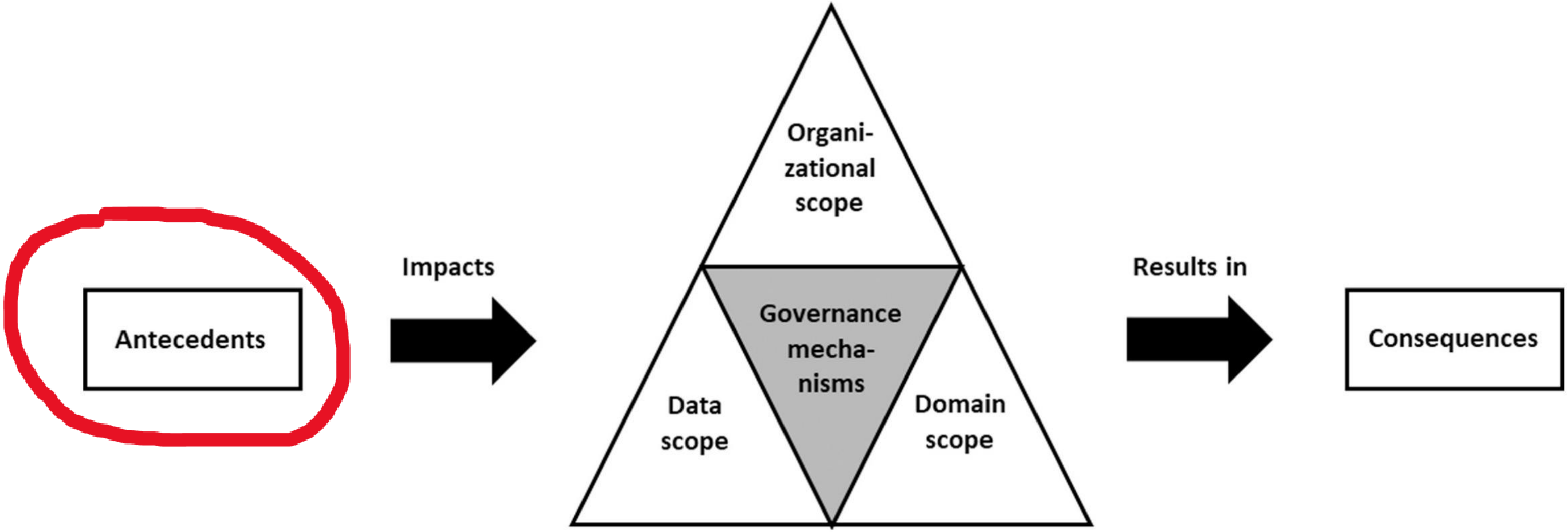
DATA GOVERNANCE: ANTECEDENTS



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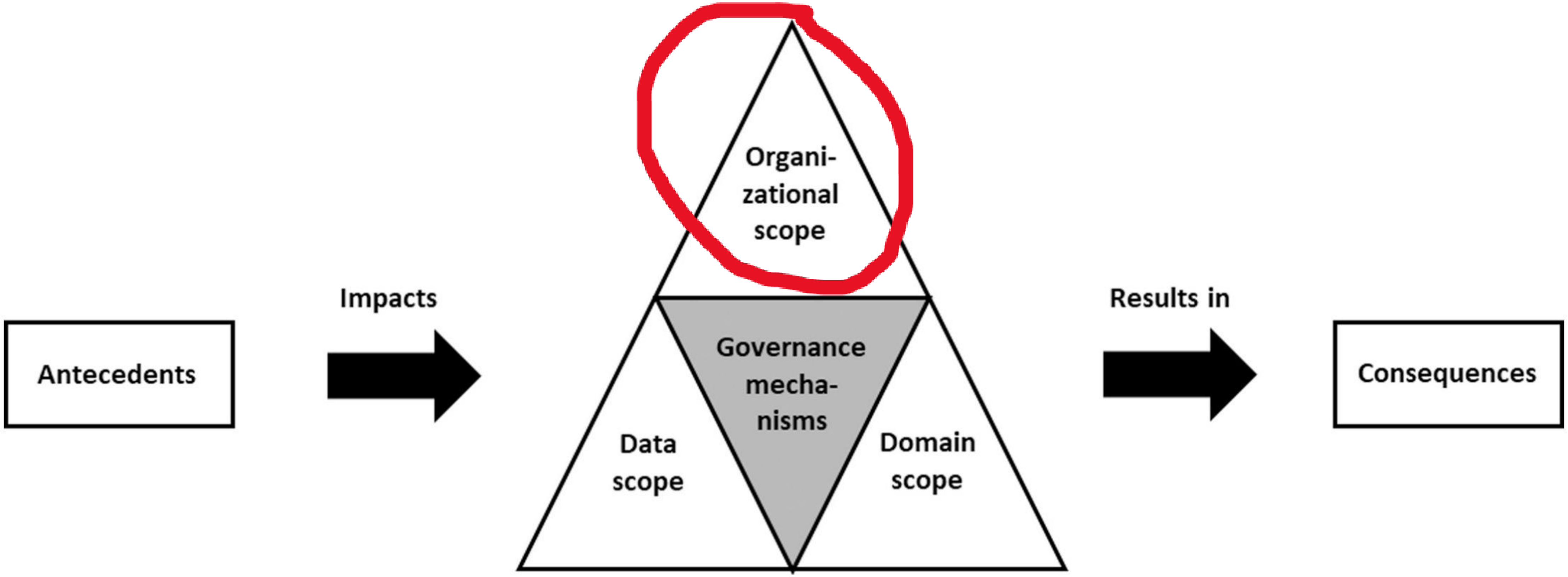
ANTECEDENTS

- Contextual environment that influence the implementation and adoption of data governance mechanisms for an organization
- External
  - Regulatory and legal environment (GDPR, DGA, HIPPA, eIDAS, etc.)
  - Market volatility
  - The country an organization is located
  - The industry or sector an organization is a part of

- Internal:
  - Organizational strategy (is data governance a part of an organization's goals?)
  - IT Strategy (centralized vs decentralized approaches)
  - Business process harmonization between units or organizations
  - IT architecture (Standardization, interoperability, monitoring systems, embedded legacy systems)
  - Organizational culture (senior management support, active leadership, etc.)

DATA GOVERNANCE: ORGANIZATIONAL SCOPE



Data governance: A conceptual framework, structured review, and research agenda. Rene Abraham, Johannes Schneider, Jan vom Brocke (2019)

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ORGANIZATIONAL SCOPE

- Intraorganizational: Data governance internal to a firm/organization/project
  - Project-based data governance focuses on data integrity

- Interorganizational: Data governance between firms/organizations/ecosystems
  - Private-Public Partnerships
  - Public to Public
  - Private to Private
  - Consortiums (Data Spaces approach)

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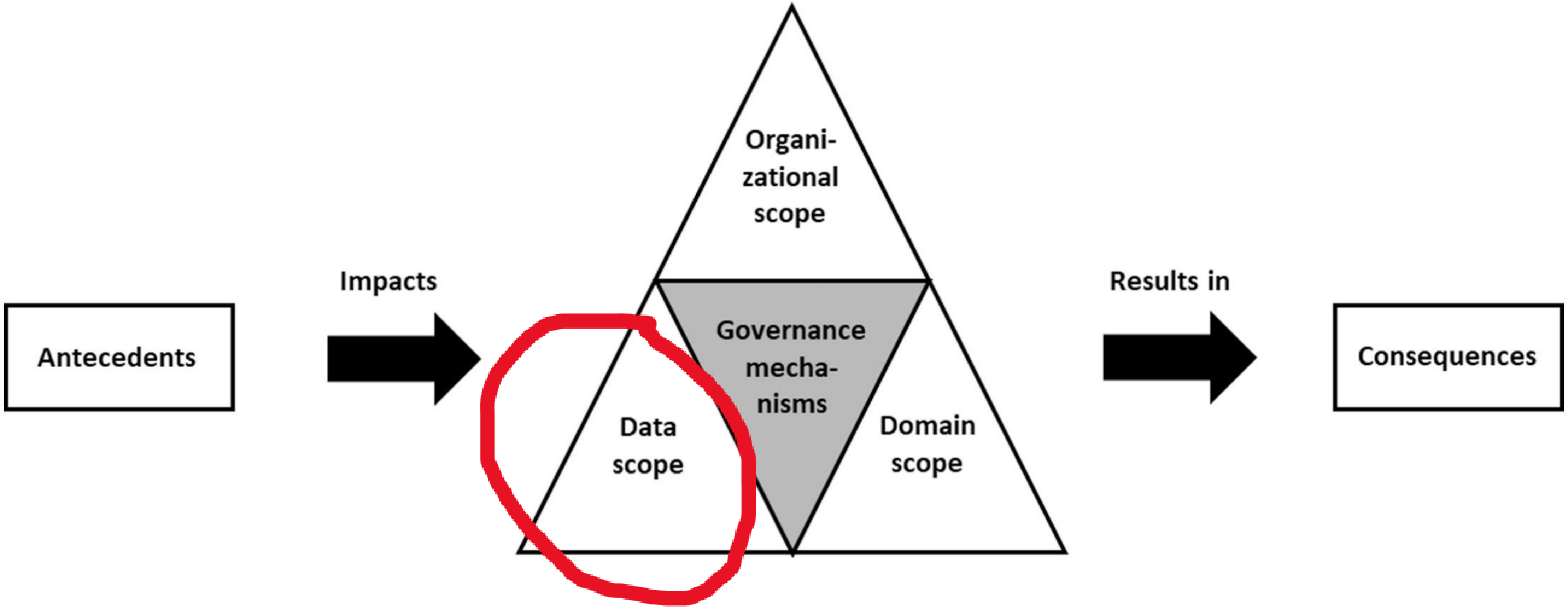
DATA GOVERNANCE: DATA SCOPE



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DATA SCOPE

Traditional Data

Master Data = Defines primary business objects of an organization
- Customer, citizen, employee, finance, patient, product, location, material, or supplier data

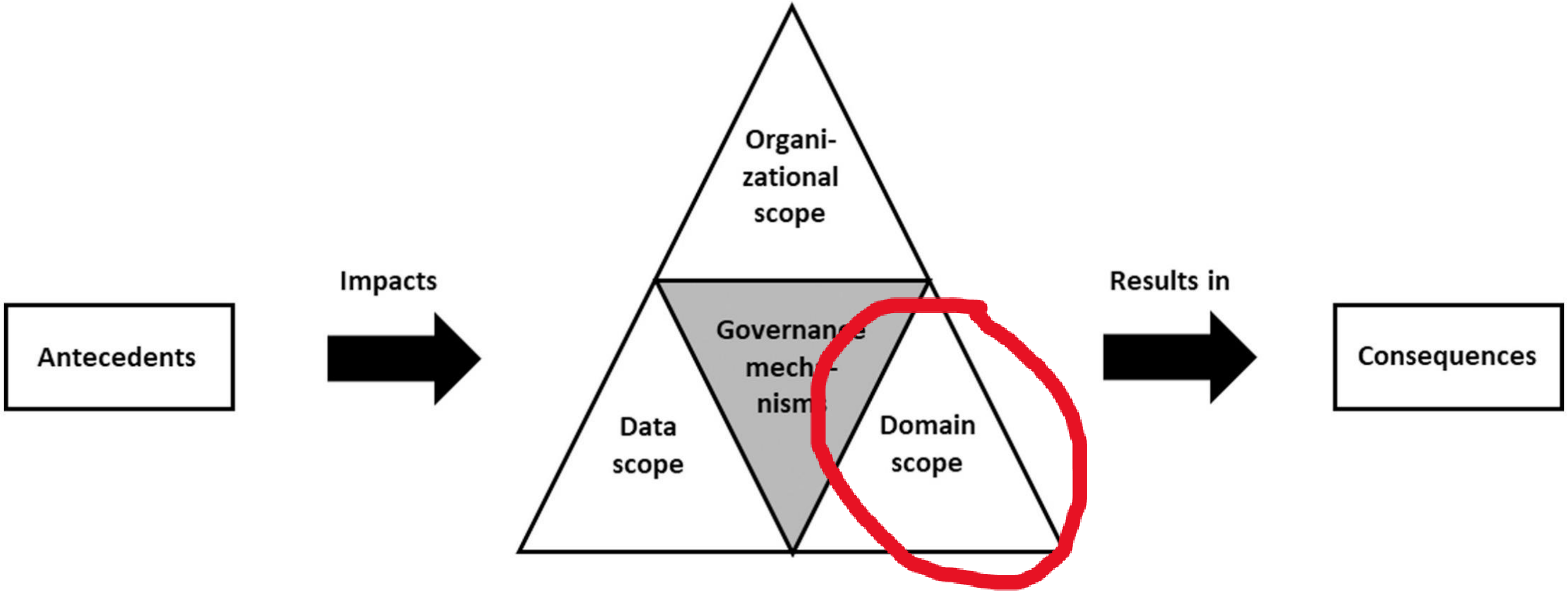
Transaction Data
Business transactions for a variety of purposes: invoices, shipments, bills, payments, etc.

Reference Data
Organizationally agreed upon standardization of data references that can be internal or external standards: i.e., ISO codes for currency, reference architectures like Data Catalog Vocabulary (DCAT) Version 3 provided by the World Wide Web Consortium

Big Data

*Web and social media data
*Machine-generated data
*Streaming data
*Biometric data

DATA GOVERNANCE: DOMAIN SCOPE



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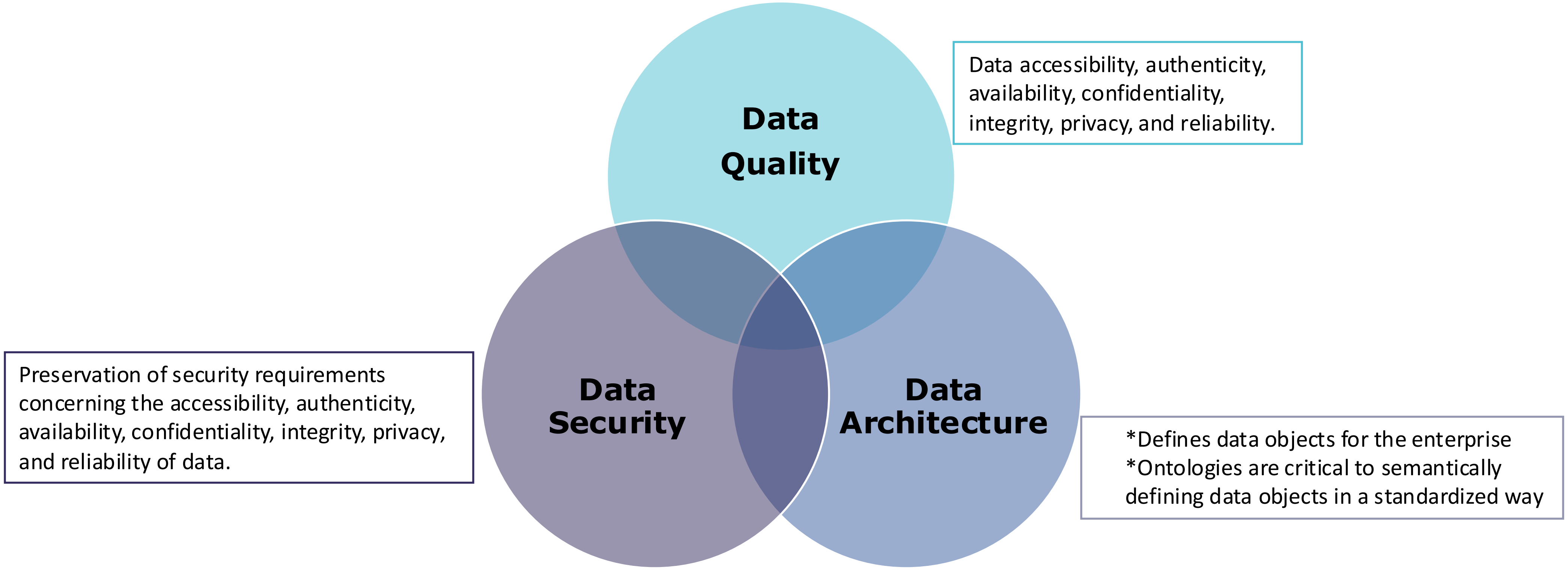
DOMAIN SCOPE

- This article considers domain scope = “data-decision domains”
- According to them, inter- or intra-organizational data governance tends to focus on a few specific data-decision domains through the following:



DOMAIN SCOPE CONTINUED

- Data lifecycle
  - defining, collecting, creating, using, maintaining, archiving, and deleting data
- Meta data (a set of data that describes and gives information about other data)
  - Data provenance
    - what source did this data come from?
    - Who created this data?
  - Meta data strategy
  - Meta data standards and management
- Data storage and infrastructure
  - focus on IT artifacts that enable effective data management across the organization
  - “must consider various hardware and software requirements such as functionality, cost, reliability, complexity, capacity, scalability, and maintainability”

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DOMAIN SCOPE CONTINUED

Data lifecycle:

defining, collecting, creating, using, maintaining, archiving, and deleting data

Metadata (a set of data that describes and gives information about other data)

- Data provenance
  - What source did this data come from?
  - Who created this data?
- Metadata strategy
- Metadata standards and management

+

Data storage and infrastructure

- Focus on IT artifacts that enable effective data management across the organization.
- "Must consider various hardware and software requirements such as functionality, cost, reliability, complexity, capacity, scalability, and maintainability

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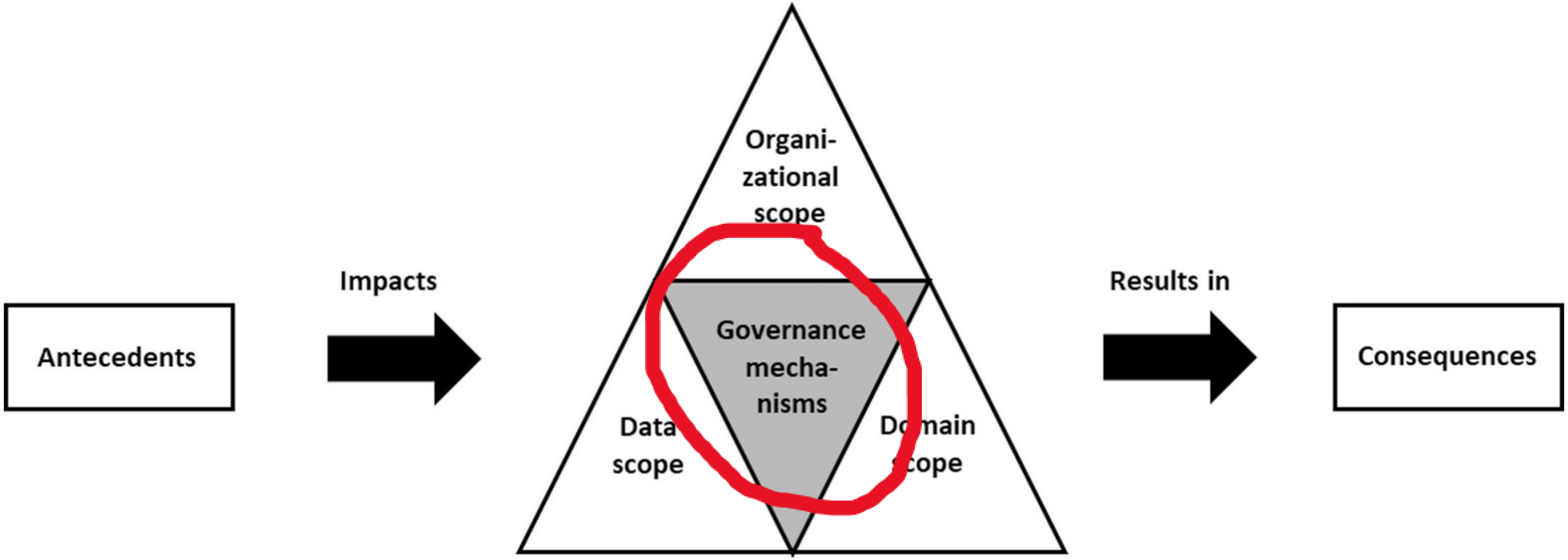
DATA GOVERNANCE MECHANISMS



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GOVERNANCE MECHANISMS

RELATIONAL MECHANISMS

- Communication and training
- Coordination of decision-making

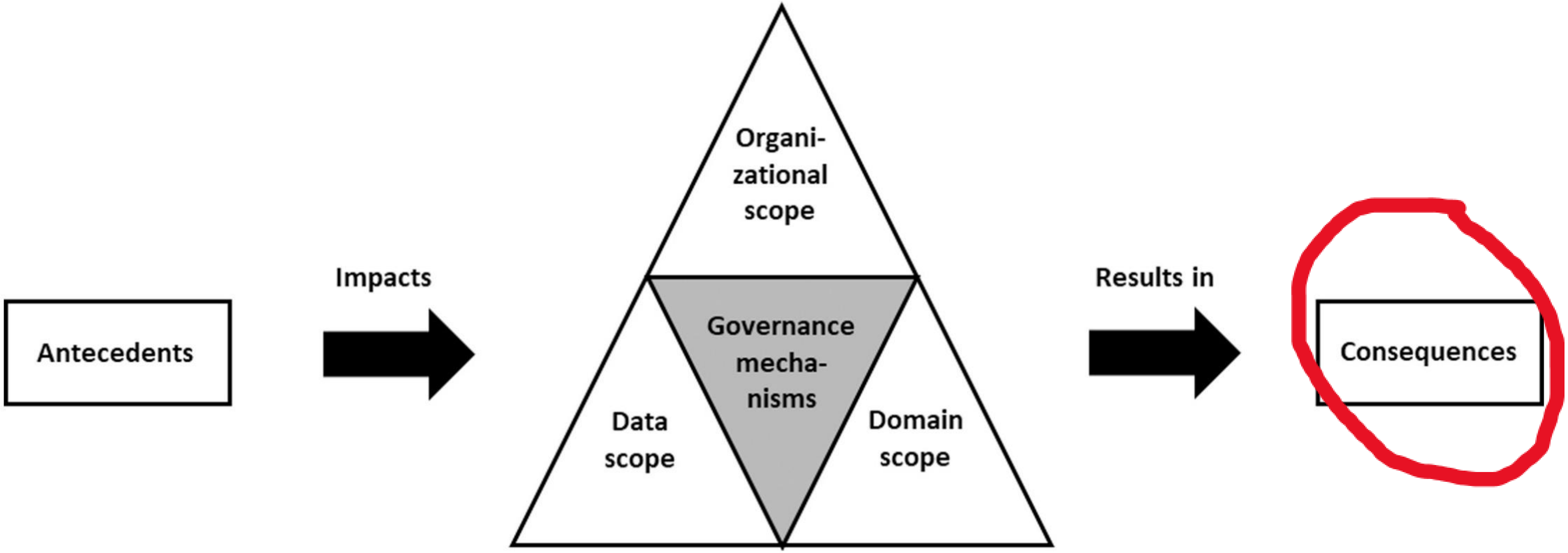
PROCEDURAL MECHANISMS

- Data strategy
- Policies, standards, processes, procedures
- Contractual agreements
- Performance measurement
- Compliance monitoring
- Issue management

STRUCTURAL MECHANISMS

- Roles and Responsibilities
- Location of decision-making authorities

DATA GOVERNANCE CONSEQUENCES



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CONSEQUENCES

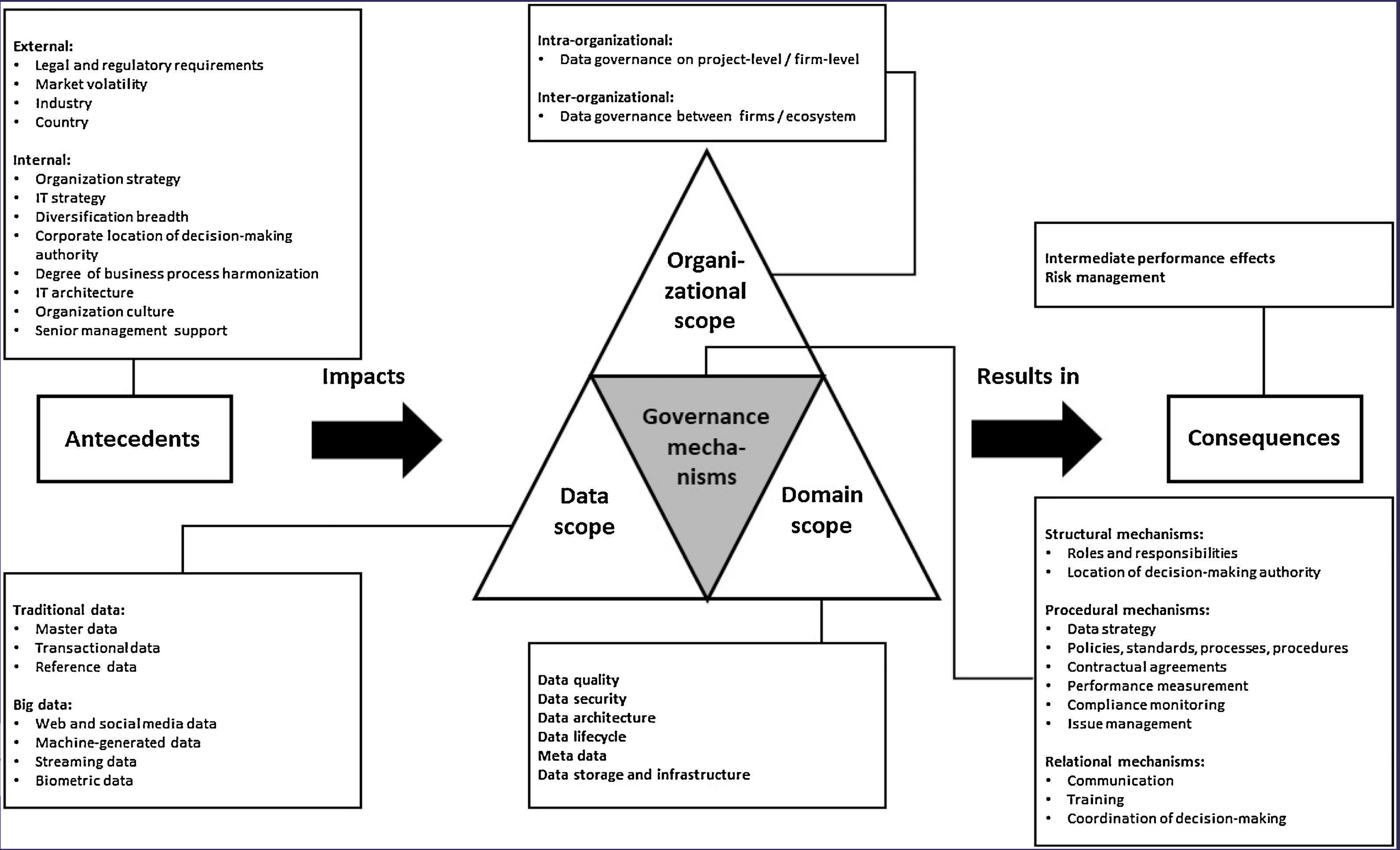
Intermediate Performance Effects (KPI driven)

- Private sector: revenue growth, cost reductions, metadata utilization, operational efficiency gains, etc.
- Public sector: hours saved, incident response time, interagency data sharing metrics, compliance scores, etc.
- Risk of "over-governance"?

Management of Data-Related Risks

Non-conformance with information policies/regulations

- Security and privacy breaches
- Data governance should help mitigate these types of risks through compliance and monitoring


External:
• Legal and regulatory requirements
• Market volatility
• Industry
• Country
Internal:
• Organization strategy
• IT strategy
• Diversification breadth
• Corporate location of decision-making authority
• Degree of business process harmonization
• IT architecture
• Organization culture
• Senior management support
Antecedents
Impacts
Intra-organizational:
• Data governance on project-level / firm-level
Inter-organizational:
• Data governance between firms / ecosystem
Organi-zational scope
Governance mecha-nisms
Data scope
Domain scope
Results in
Intermediate performance effects
Risk management
Consequences
Traditional data:
• Master data
• Transactional data
• Reference data
Big data:
• Web and social media data
• Machine-generated data
• Streaming data
• Biometric data
Data quality
Data security
Data architecture
Data lifecycle
Meta data
Data storage and infrastructure
Structural mechanisms:
• Roles and responsibilities
• Location of decision-making authority
Procedural mechanisms:
• Data strategy
• Policies, standards, processes, procedures
• Contractual agreements
• Performance measurement
• Compliance monitoring
• Issue management
Relational mechanisms:
• Communication
• Training
• Coordination of decision-making

GROUP REFLECTION: Is there anything missing from

Data governance: A conceptual framework, structured review, and research agenda. Rene Abraham, Johannes Schneider, Jan vom Brocke (2019)?

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GROUP REFLECTION: In your work experience, has your organization ever struggled with Data Governance?

What were some of the challenges you faced?

OPEN DATA AND TRANSPARENCY

DR. ERIC BLAKE JACKSON

LECTURER, NEXT GEN DIGITAL STATE UNIVERSITY

TALTECH UNIVERSITY

WHAT'S SO SPECIAL ABOUT OPEN DATA?

- Open Data is unique because of its licensing, accessibility, reusability, and public nature (Digital Public Good)
- Very context-dependent on data quality, national/local policies, domains
- Needs to be interoperable: aka machine-readable formats
- Open Data can:
  - Be used for generating new in cross-domain insights/knowledge
  - Help build new public and private services
  - Facilitate transparency through publishing procurements, contracts, budgets, etc.
  - Be used for data-driven decision-making and now AI development
  - Public Sector tends to use online portals as an interface

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THEORETICAL MODEL OF OPEN DATA (SAFAROV ET AL., 2017)

16 I. Safarov et al. / Utilization of open government data

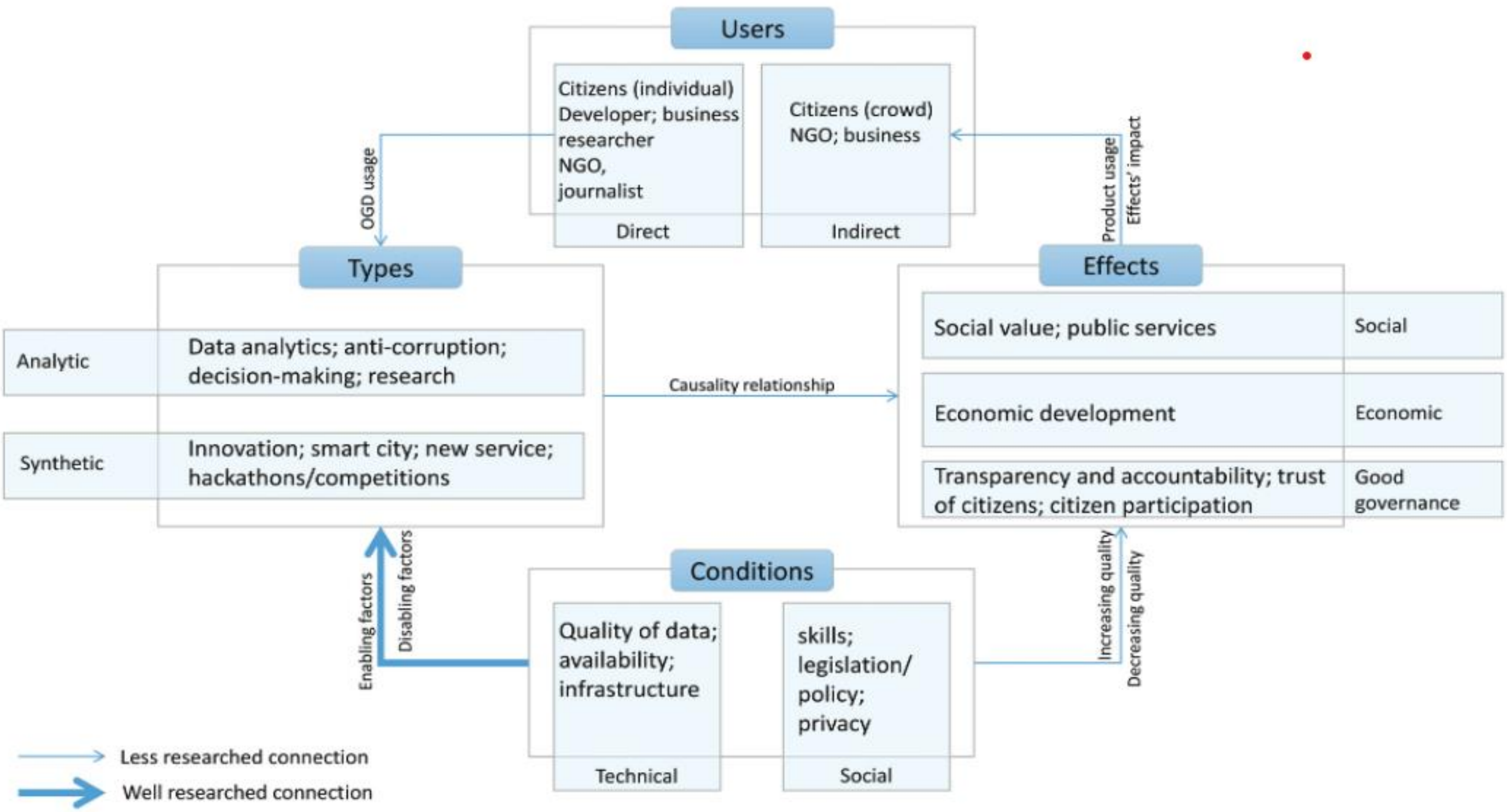

Fig. 9. OGD utilization framework.

TRENDS ON OPEN DATA

- OECD 2023 OURdata Index
- Covers 40 countries with a 3 pillar and 9 sub-pillar methodology

Table 1. Structure of the OURdata Index

Composite:	OURdata Index		
3 pillars	1. Data availability	2. Data accessibility	3. Government support to data-reuse
9 sub-pillars:	1.1. Content of the open by default policy	2.1. Content of the free and open access to data policy	3.1. Data promotion initiatives and partnerships
	1.2. Stakeholder engagement for data release	2.2. Stakeholder engagement for data quality and completeness	3.2. Data literacy programmes in government
	1.3. Implementation (Availability of high-value datasets)	2.3. Implementation (Accessibility of high value datasets)	3.3. Monitoring impact

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OPEN DATA TRENDS

- “The best performing countries in the 2023 OURdata Index are Korea, France, Poland, Estonia, Spain, Ireland, Slovenia, Denmark, Sweden, and Lithuania”
- “Only 48% of high-value datasets are available as open data across OECD countries.”
- “While the COVID-19 pandemic positively influenced the publication of open health data, only 43% of datasets in this category are available today.”
- “High-value datasets related to government finances and public accountability are rarely available.”
- “OECD countries have improved the quality of open government data, an important capability considering recent advancements in Artificial Intelligence (AI)”

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ESTONIA'S OPEN DATA PORTAL:

HTTPS://ANDMED.EESTI.EE/

MY EXPERIENCE WITH OPEN DATA

- Open Data was a hot topic in the late 2010s
- Was framed as a panacea, but not fulfilling these expectations
- Fell a bit out of favor, but with COVID-19 and AI, open data is back as a trend (imo)
- Governments are reluctant to open their datasets because it is a valuable asset and adverse to transparency
- EU sees open data as a key facilitator of its data policies/strategies
- Who are the actual end-users/beneficiaries?

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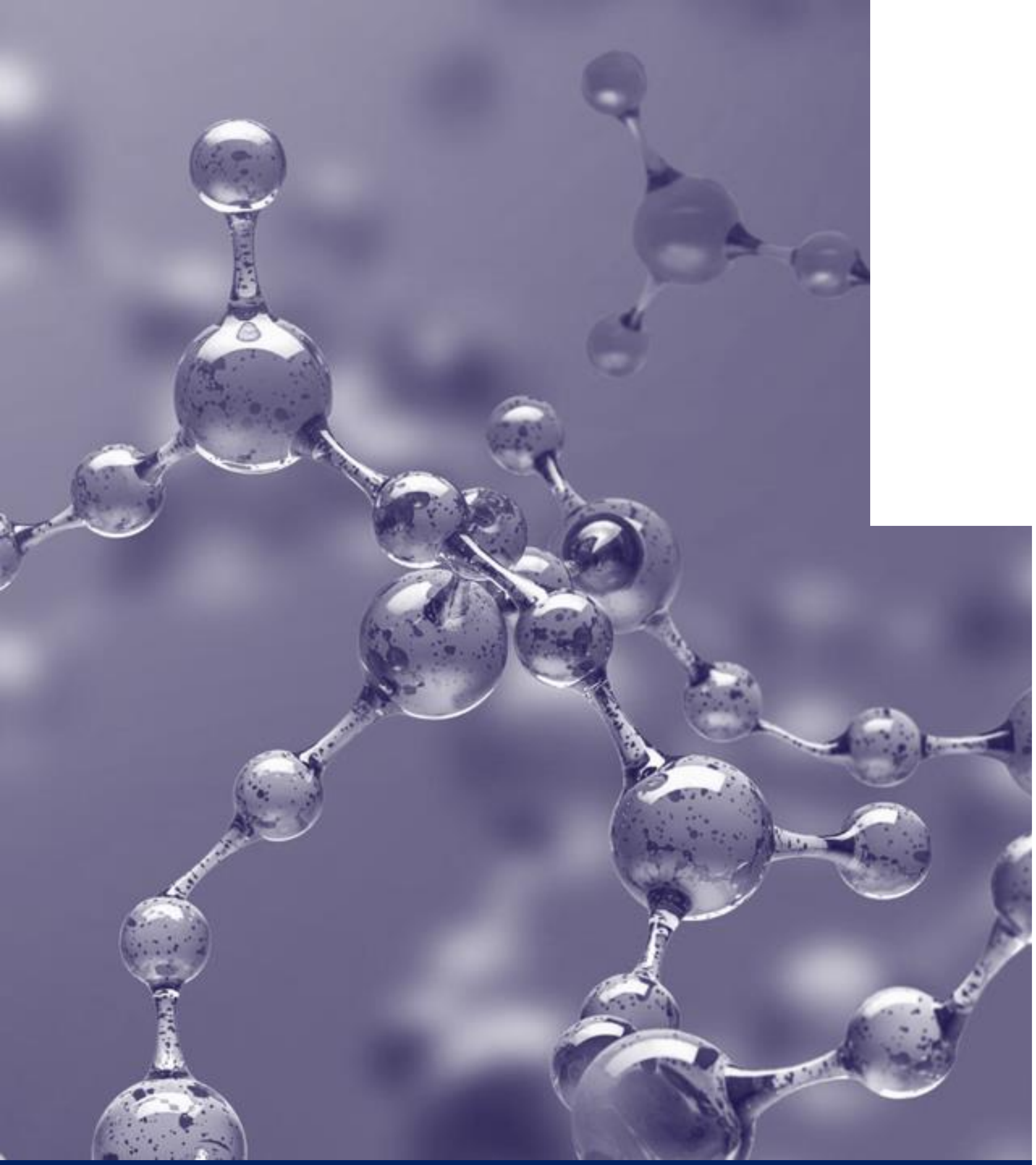
HAVE YOU EVER USED OPEN DATA BEFORE IN YOUR WORK OR PERSONAL LIFE?



THE FUTURE OPERATING MODEL OF THE DIGITAL STATE

GROUP SENTIMENT REFLECTION

When you think about artificial intelligence, what feelings come to mind first — excitement, curiosity, worry, skepticism, or something else? Why?

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GROUP SENTIMENT REFLECTION

Do you feel AI is more of an opportunity or a risk for society as a whole? What influences your view?

GROUP SENTIMENT REFLECTION

In your view, what is the most *promising* use of AI in government — and what is the most *concerning*?

GROUP SENTIMENT REFLECTION

How comfortable would you feel if an AI system made or supported decisions that affect citizens (for example, social benefits, permits, or healthcare services)?

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What the hell is AI?

- AI has been around since the 1950s/60s, but fell out of favor until the early 2000s
    - 1955 the first mention of "Artificial Intelligence" at a workshop held at Dartmouth University

- *"Artificial Intelligence (AI) refers to the simulation of human intelligence in machines programmed to perform tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, perception, understanding natural language, and even interacting with the environment. AI systems are designed to mimic cognitive functions, allowing them to adapt and improve performance based on experience"* ChatGPT definition

- THE APPLICATION OF AI IS NOT ONE SINGULAR THING!

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AI General Concepts

- **Narrow or Weak AI:**
- **Definition:** Weak AI is designed and trained for a particular task or a narrow set of tasks. It operates within a limited context and cannot perform tasks beyond its predefined scope.
- **Example:** Virtual personal assistants like Apple's Siri or Amazon's Alexa, which are specialized in voice recognition and responding to user commands but do not possess general intelligence.

- **General or Strong AI:**
- **Definition:** Strong AI refers to a system with the ability to understand, learn, and apply knowledge across a wide range of tasks at a human level. It can perform any intellectual task that a human being can.
- **Example:** Currently, there are no examples of strong AI in existence. It represents a theoretical concept of highly advanced AI that exceeds human intelligence.

- **Artificial Narrow Intelligence (ANI):** Same as Weak AI, specialized in a specific task.(ChatGPT)
- **Artificial General Intelligence (AGI):** Equivalent to Strong AI, possessing human-like cognitive abilities across various domains.
- **Artificial Superintelligence (ASI):** An advanced form of AI that surpasses human intelligence in every aspect, potentially leading to unprecedented capabilities.

AI In Application

1. **Machine Learning (ML):**
    1. ML is a subset of AI that involves the use of algorithms and statistical models to enable a system to improve its performance on a task without explicit programming. It focuses on learning patterns and making predictions.
2. **Natural Language Processing (NLP):**
    1. NLP involves the interaction between computers and humans using natural language. It enables machines to understand, interpret, and generate human-like text or speech.
3. **Computer Vision:**
    1. This involves giving machines the ability to interpret and make decisions based on visual data. Computer vision is used in facial recognition, image and video analysis, and other visual tasks.
4. **Speech Recognition:**
    1. AI systems equipped with speech recognition can understand and interpret spoken language, converting it into text or performing specific actions based on verbal commands.
5. **Expert Systems:**
    1. Expert systems are AI programs designed to mimic the decision-making abilities of a human expert in a particular domain. They use knowledge bases and inference engines to provide solutions or make decisions.
6. **Robotics:**
    1. AI in robotics involves programming machines to perform tasks in the physical world, such as object manipulation, navigation, and interaction with the environment.

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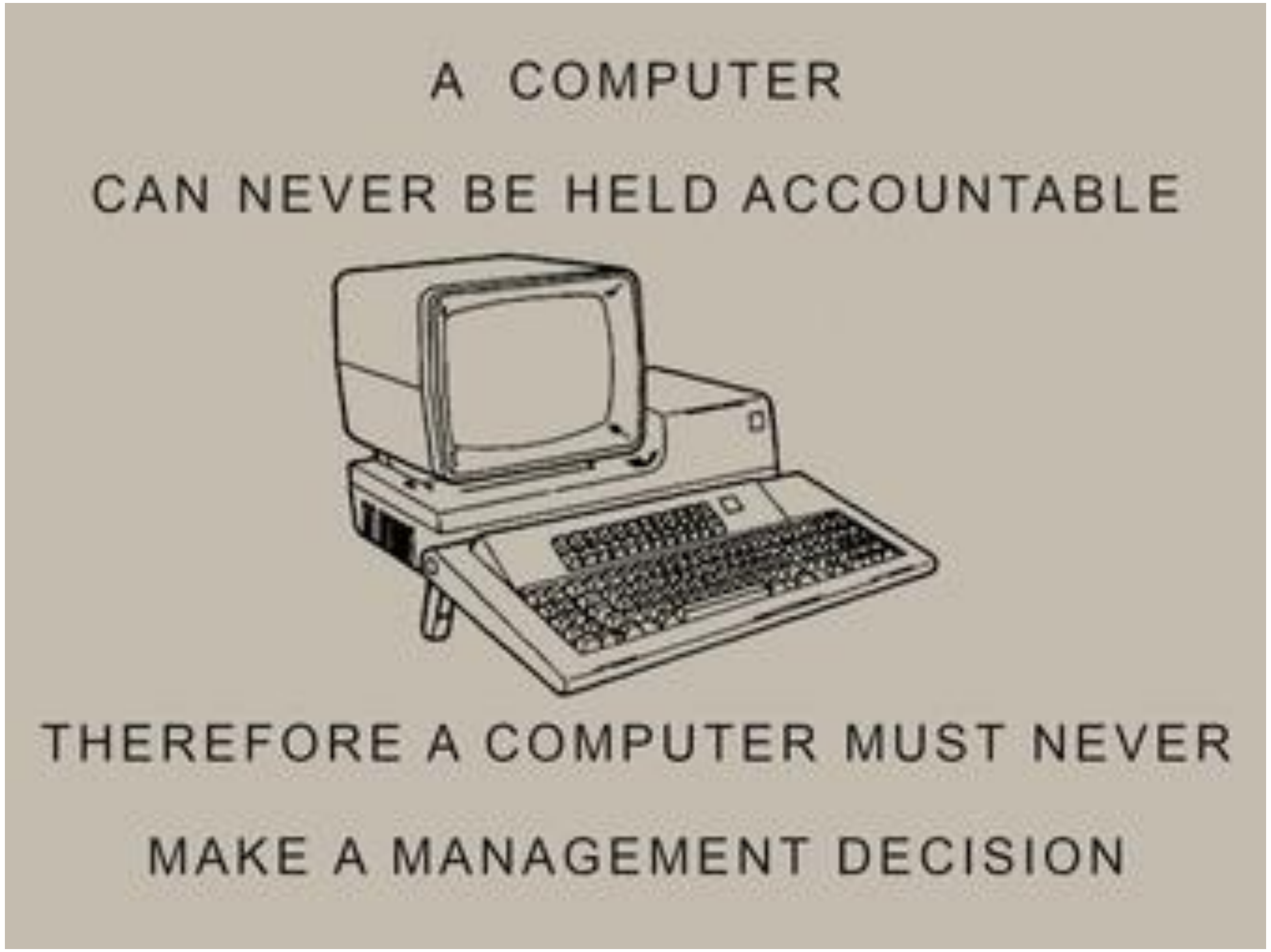

1970, IBM Manual

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“THE AGENTIC STATE: HOW AGENTIC AI WILL REVAMP 10 FUNCTIONAL LAYERS OF PUBLIC ADMINISTRATION”

- Visionary White Paper published recently about the impact AI will have on the digital state (Berlin Global Government Technology Centre)
- **Premise:** Public governance in high income countries is failing, despite the massive amount of resources poured into bureaucratic functions
- Low-income countries must do more with a lot less, “leapfrogging”
- AI is developing rapidly and intensively
- Governments are conservative by nature, just starting to dip their toes into the AI pool

Ilves, L., with Kilian, M., Peixoto, T., Velsberg, O. (2025). *The Agentic State: How Agentic AI Will Revamp 10 Functional Layers of Government and Public Administration. Version 1.0, May 2025.*

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"THE AGENTIC STATE: HOW AGENTIC AI WILL REVAMP 10 FUNCTIONAL LAYERS OF PUBLIC ADMINISTRATION"

- Generative AI -→ AI Agents

- "AI Agents will be systems that gather, process, synthesize information, execute tasks and automation, engage with humans or other agents"

- "Systems of agents, what we refer to as agentic AI, are predicted to take over tasks that previously required significant human oversight and judgement, such as software engineering, IT operations, human resources, finance, customer service, marketing, operations, and supply chain management"

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"THE AGENTIC STATE: HOW AGENTIC AI WILL REVAMP 10 FUNCTIONAL LAYERS OF PUBLIC ADMINISTRATION"

This whitepaper breaks down the broad notion of an *Agentic State* into ten distinct functional layers of government, ordered across three dimensions:

Operations	1. Service Delivery and User Experience 2. Internal Workflows 3. Data Governance, Management, and Operations 4. Crisis Response and Resilience
Regulation and Governance	5. Compliance and Reporting 6. Policy and Rulemaking
Foundations	7. Leadership 8. Workforce and Culture 9. Tech Stack 10. Public Procurement

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SERVICE DELIVERY AND USER EXPERIENCE

- Personalization and multi-channel interaction
  - Universal accessibility linguistically and channels/devices
  - Emotionality can be customized to interaction context

- Proactive e-services
  - Moving from static “life event” based proactivity towards personalized and contextual proactivity

- Defined rules + resources = intent recognition and full-service provision

DATA GOVERNANCE, MANAGEMENT, AND OPERATIONS

- In my work, public sector struggles tremendously with data governance and handling Big Data sets: time constraints, data quality, who owns the data?

- Agents can help with:
  - Unique identities (generation and provenance)
  - Metadata (data lineage)
  - Transparency
  - Managing data as a product
  - Agents as data scientists

COMPLIANCE AND REPORTING

European Union = overregulation to the point compliance activities is a major economic segment of GDP (3.5 percent)

Choking innovation, and is a lot of paperwork for both sides

Thoroughness: Agents can examine the full picture of enterprise data, gigabits per second streams, to discover risks.

Managing complexity: AI can act as compliance copilots. They can ingest new rules and regulations, map them to enterprise systems, and highlight where necessary changes, exceptions, or trade-offs are needed.

Real-time validation: Instead of depending on quarterly reports or scheduled inspections, a statement of conformity becomes a live reflection of the present.

ESTONIAN AI: BUROKRATT

Estonian Public Sector and AI

- The Estonian government has been implementing AI for service provision and data-driven decision-making for quite some time now.
- AI solutions have been implemented in the public sector of Estonia around 120 times
- Around 60 public authorities have implemented projects with an AI component to improve the efficiency of their work
- https://www.kratid.ee/ai-use-cases
- **Bürokratt**

Bürokratt – a virtual assistant, thanks to which citizens no longer have to spend a single extra minute communicating with the state.

- ✓ All answers in one place
- ✓ Reliable information
- ✓ Available 24/7
- ✓ The user can choose the communication channel (chat, SMS, email, phone, voice assistant)
- ✓ Free of charge

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